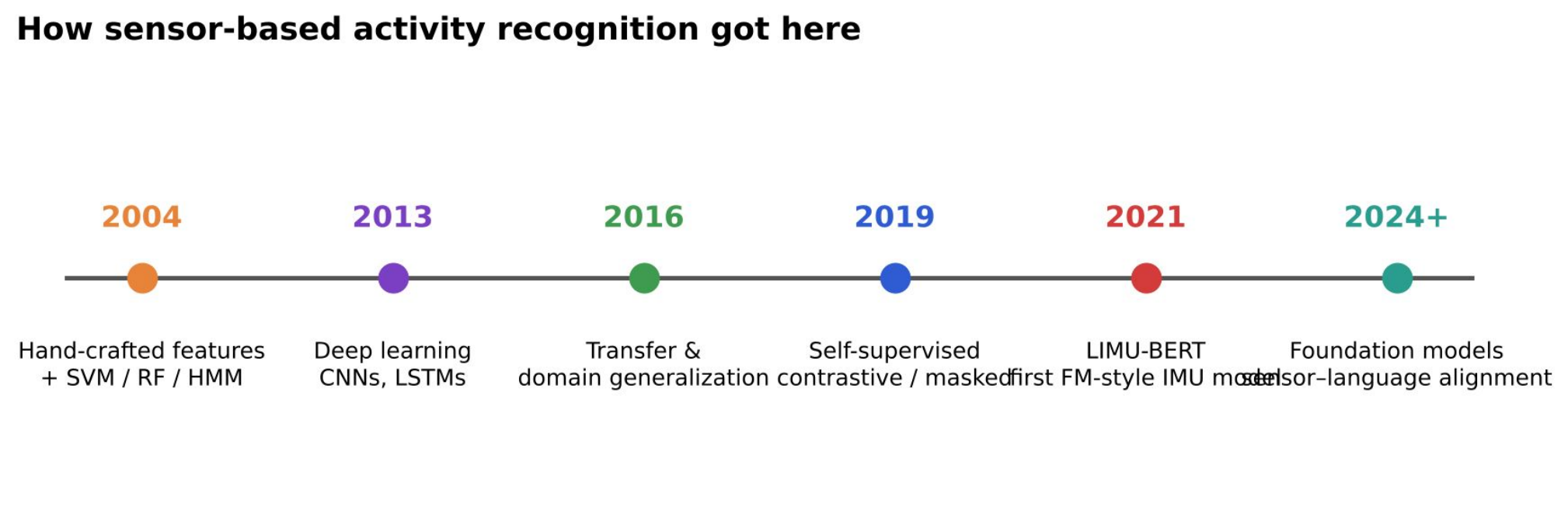
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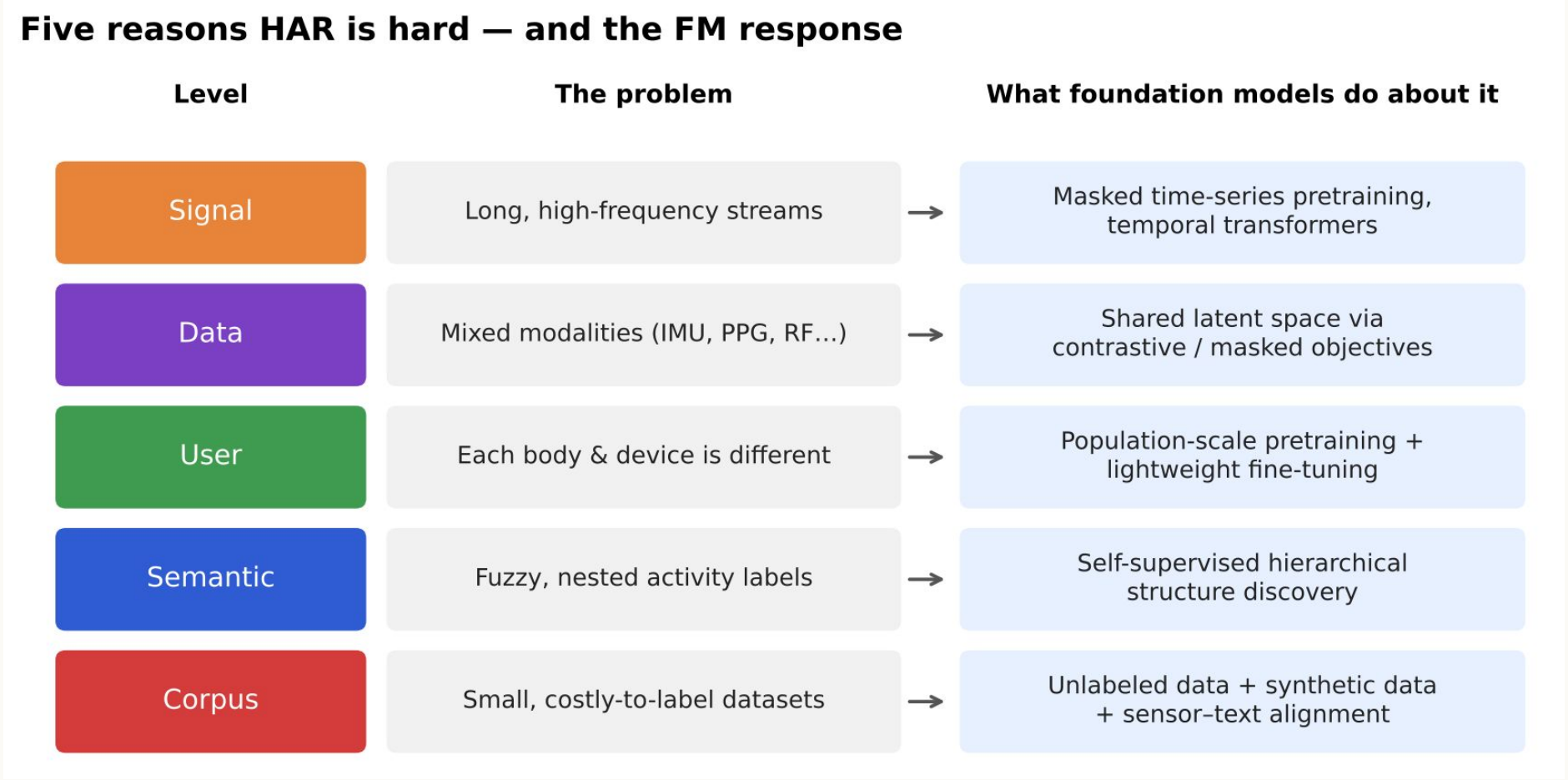
Foundation Models for Sensor-based Activity Recognition

Twenty years in one line

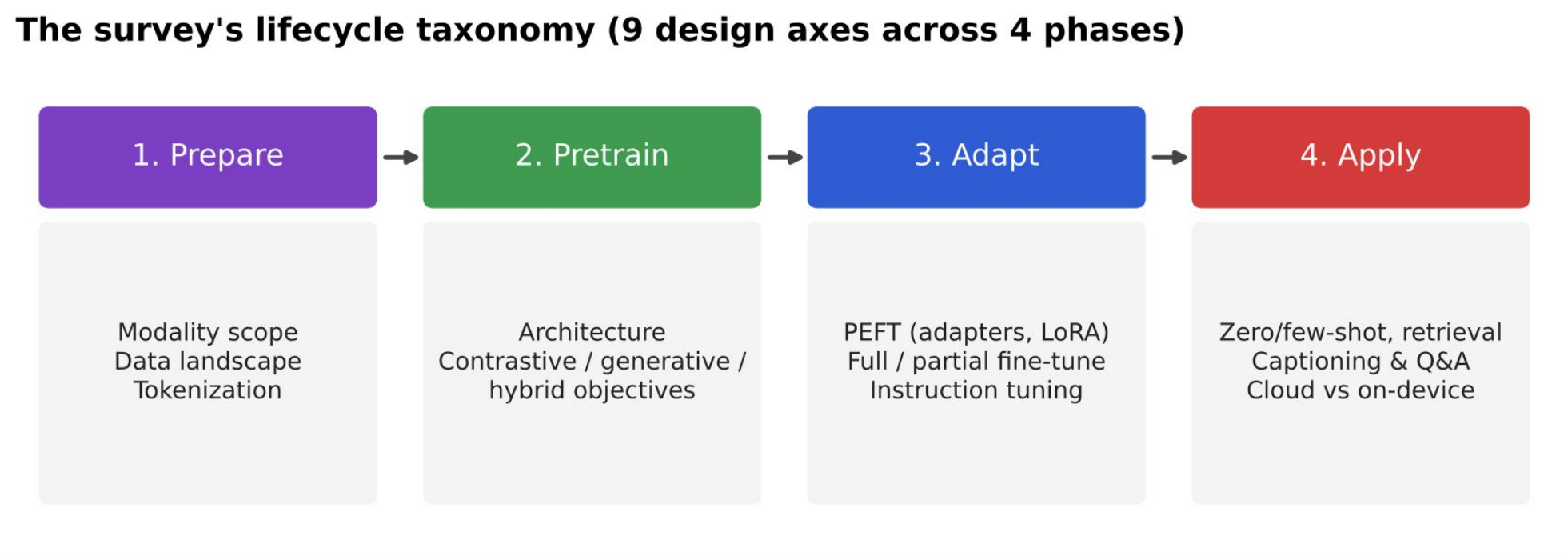


HAR followed vision and NLP: features → deep nets → self-supervised → **pretrain once, reuse everywhere**.

Why HAR is hard

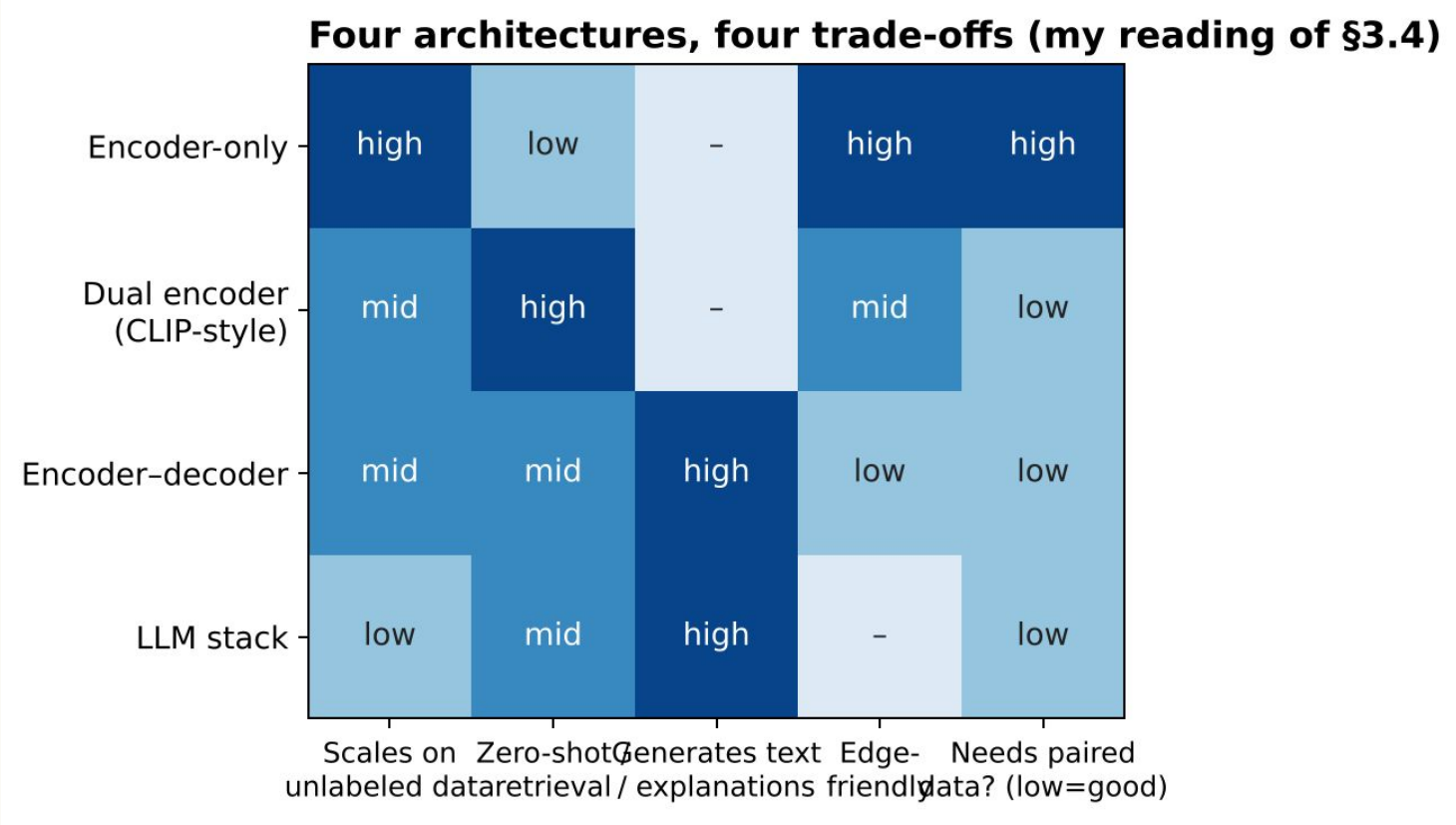


The survey’s map: a four-phase lifecycle



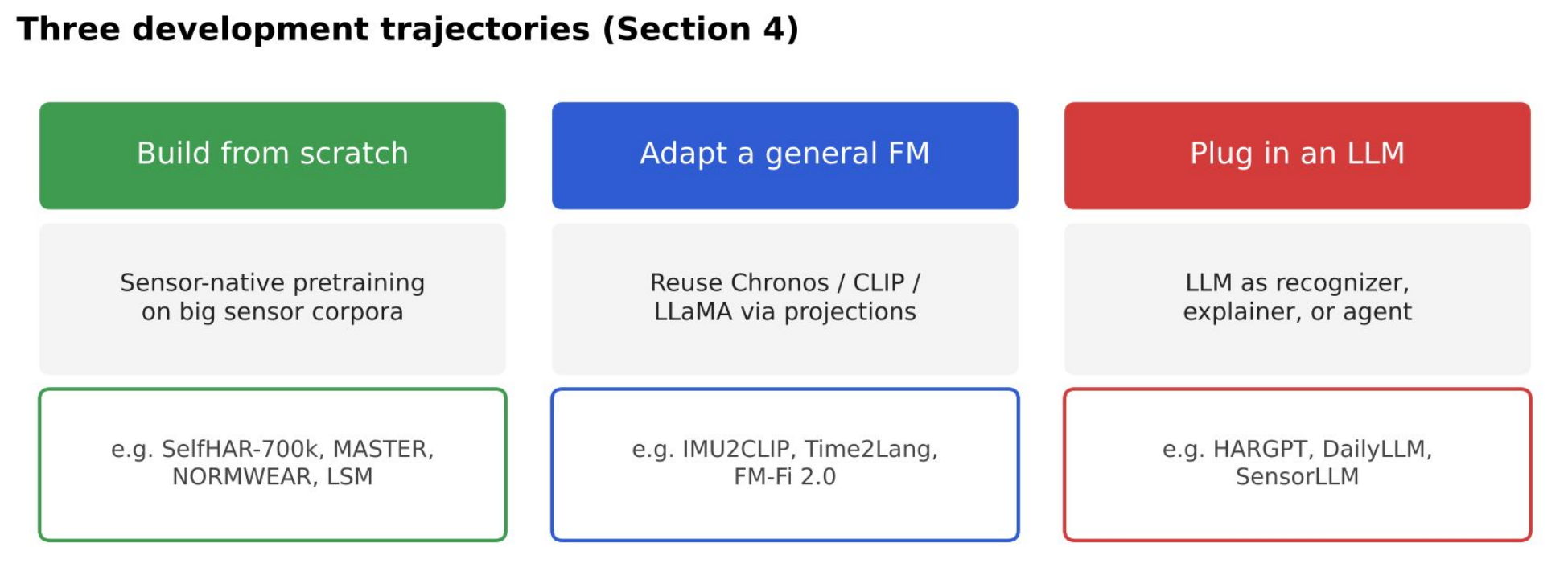
Nine design axes, from tokenization to deployment. **Hybrid pretraining** (masked + contrastive) and **LoRA-style adaptation** are the current defaults.

Four architectures, four trade-offs



Ratings are my reading of the survey's Section 3.4, not numbers from the paper.

Three ways to build one



LSM shows wearable models follow **scaling laws**, but only when data grows with model size.

Match every claim to a metric

Capability	Protocol	Metric
Cross-user / device	Train on A, test on unseen B	F1 under shift
Zero-shot / open-set	Unseen activity classes	Open-set F1
Retrieval	Sensor ↔ text / video	Recall@K, mAP
Generative	Held-out spans	Reconstruction error
On-device	Real hardware	Latency, energy

Accuracy on a random IID split says little about a model meant to generalize.

My two cents

What works

- Lifecycle taxonomy beats a model list
- Evaluation treated as a first-class problem
- Clear scope: no vision-based HAR

What I’d push on

- Radar scores are subjective
- “Foundation model” is stretched
- Smart-home event data barely covered
- No unified benchmark numbers

TAKEAWAY

The next leap in wearables comes from unlabeled data, not bigger labeled datasets.

Sensor-native pretraining + light adaptation + LLM reasoning will likely converge into one model.